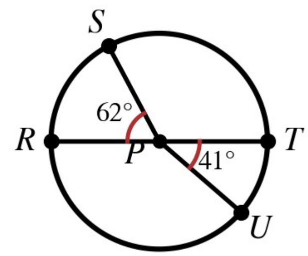
Speak **L**ike **A** **M**athematician:

<u>central angle:</u>

<u>minor arc / major arc:</u>

<u>semicircle:</u>



Arc Measure Rules

The measure of a minor arc equals the measure of its ____________________ angle.

A major arc measures 360° minus its ____________________ arc; a semicircle measures exactly _______ .

Arc Addition Postulate: m$\widehat{ABC}$ = m$\widehat{AB}$ + m ____________ .

Example 1 – Arcs in Circle P

RT is a diameter, m$\widehat{RS}$ = 62° and m$\widehat{UT}$ = 41° (figure top right).

a) Find m$\widehat{ST}$.

b) Find m$\widehat{RU}$.

c) Find m$\widehat{STU}$.

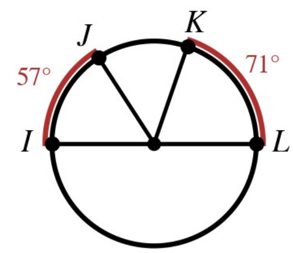
d) Find the major arc m$\widehat{SRU}$.

Example 2 – Diameter IL

IL is a diameter, m$\overset{\frown}{IJ}$ = 57° and m$\overset{\frown}{KL}$ = 71°.

a) Find m$\overset{\frown}{JK}$.

b) Find m$\overset{\frown}{LIJ}$.

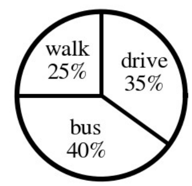


Example 3 – Survey Circle Graph

A class survey: 25% walk, 40% ride the bus, 35% drive.

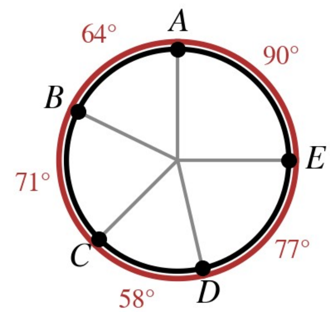
a) Find the central angle for each category.

b) Check your three angles.



Example 4 – Five Arcs

In the circle at the right, $m\widehat{AB} = 64°$, $m\widehat{BC} = 71°$, $m\widehat{CD} = 58°$, $m\widehat{DE} = 77°$, $m\widehat{EA} = 90°$.



a) Find $m\widehat{BCD}$.

b) Find $m\widehat{EAB}$.

c) Verify that all five arcs make a full circle.

Example 5 – Congruent or Just Similar?

a) A 65° arc is drawn in a circle of radius 4 and in a circle of radius 7. Are the arcs congruent?

b) When ARE two arcs congruent?

c) Why are any two circles similar?